ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C3 · HEAT

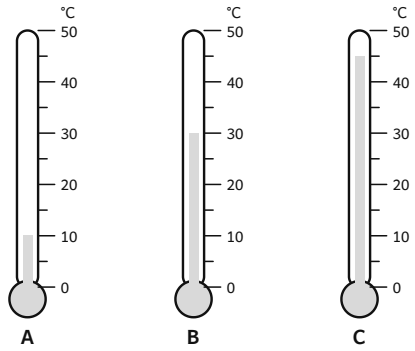
Hot and cold facts

★☆☆
Grade 3

Learning goals

- Read thermometers.
- Sort changes.

1 Read the thermometers. [3]



A	B	C
10 °C	30 °C	45 °C

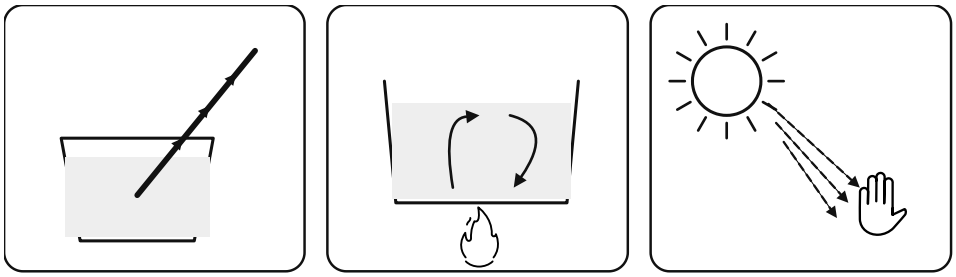
2 Is heat added or removed? [6]

SORT THESE

rice cooks **ice cream melts** **water freezes in the freezer**
melted wax goes hard **wet clothes dry in the sun** **hot tea cools**

Heat added	Heat removed
ice cream melts	water freezes in the freezer
rice cooks	hot tea cools
wet clothes dry in the sun	melted wax goes hard

3 **Name the way heat travels in each picture.** [3]



A B C

A	B	C
conduction	convection	radiation

ANSWER KEY

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THINK IT THROUGH · C3 · HEAT

Thinking about heat

★★☆
Grade 3

Learning goals

- Explain conductors and insulators.

1 Why do cooking pots have wooden or plastic handles? [2]

Wood and plastic are insulators, so the handle does not get too hot to hold.

2 Why do we wear several layers of clothes in winter? [2]

Air is trapped between the layers; air is an insulator, so it keeps our body heat in.

3 Sort the materials. [5]

SORT THESE

steel spoon **aluminium pot** **wool** **wooden spoon** **plastic handle**

Conductor	Insulator
steel spoon	wooden spoon
aluminium pot	wool
	plastic handle

ANSWER KEY

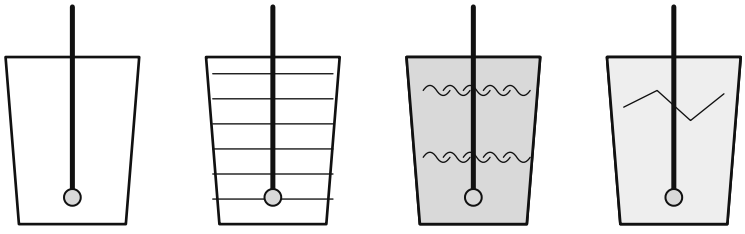
INVESTIGATE · C3 · HEAT

Which cover keeps tea hot the longest?

★★☆ Grade 3

Learning goals

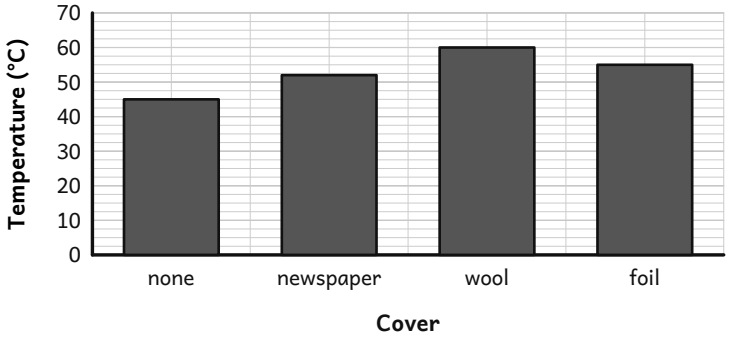
- Measure temperature.
- Draw a bar chart and conclude.



Class 3A filled four cups with hot water at **80 °C**. They wrapped three cups in different materials. After 20 minutes they measured the temperature. (Only an adult should pour the hot water.)

1 **Draw a bar chart.** [3]

Cover	none	newspaper	wool	foil
Temperature after 20 min (°C)	45	52	60	55



2 **Which cover was the best insulator?** [1]

wool

3 **Why did the cup with no cover cool fastest?** [2]

Nothing stopped the heat escaping into the air.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C3 · HEAT

Keeping warm safely in winter

★★★
Grade 3

Learning goals

- Use what you know about heat to stay safe.

Science | Health | Community

In winter, nights in the north of Bangladesh can get very cold, below **10 °C**. Many families sit around small open fires to keep warm.

Every winter, some people are badly burnt when loose clothes such as a sari, orna or shawl catch fire. Doctors and fire fighters teach families how to keep warm safely: wear warm layers and use quilts, sit at a safe distance from fires, tie back loose clothes, and never leave a fire alone.

Based on winter safety advice in Bangladesh.

1 **How does a fire warm people who are sitting near it?** [1]

by radiation: heat rays travel from the fire

2 **Write three safety rules for sitting near a fire.** [3]

sit at a safe distance; tie back loose clothes; never leave the fire alone; keep water or sand nearby

3 **If someone's clothes catch fire, they should stop, drop and roll. Draw the three steps.** [3]

Stop, drop, roll

stop moving; drop to the ground; roll over to put out the flames.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C3 · HEAT

Show what you know

★★★
Grade 3

Learning goals

- Use heat words.

1 Find the words.

G	O	F	C	O	O	L	G	H
H	Y	P	W	D	G	R	J	Y
D	S	U	N	U	R	U	S	L
N	N	W	A	L	T	D	C	D
J	B	K	W	H	M	E	L	T
B	H	F	R	E	E	Z	E	P
O	J	F	I	R	E	F	Y	H
I	C	G	H	E	A	T	K	G
L	G	G	T	F	D	N	P	E

boil cool fire freeze heat melt sun

2 Temperature is measured with *a*... [1]

◯ ruler
● **thermometer**
◯ magnet

3 Heat travelling up *a* metal spoon is... [1]

● **conduction**
◯ convection
◯ radiation

4 Challenge. Design *a* box to keep *a* lunch of hot rice warm. What materials would you use? [2]

a metal tiffin box wrapped in wool or cloth, inside a plastic bag; insulators trap the heat